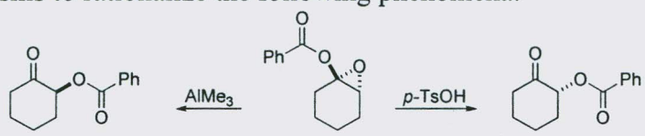
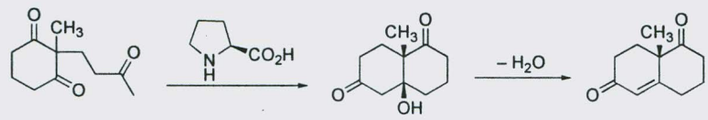
Organic Cume
April 1, 2011

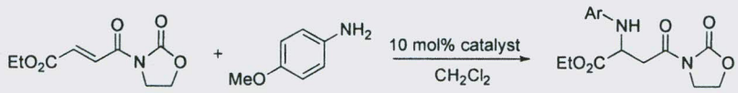
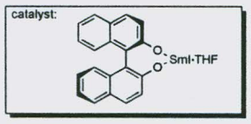
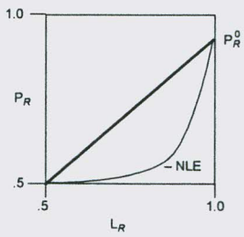
1. Draw mechanisms to rationalize the following phenomena:



2. Many people credit the advent of what is now called *organocatalysis* to the Hajos-Parrish synthesis of the Wieland-Mischer ketone, shown below. Propose a mechanistic rationale for this reaction, which affords the correct absolute configuration.

Hajos and Parrish (1974):



Wieland-Mischer ketone

3. The reaction below exhibits a *negative* nonlinear effect. Provide a rationale for the negative nonlinear effect. In the graph, P_R and L_R represent the fraction of the *R* enantiomer of the product and the ligand, respectively. (You do not need to know the mechanism of this reaction to answer the question.)